# An embedding into an n-type has an n-type domain

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author thm environment at source lines 102-105 and its proof environment opens on line 106, the next line; the pairing is by adjacency and the retained proof range 106-109 is the whole environment.

*Editorial note.* The retained counter values (chapter 7, section 1, theorem offset 5) make the counted theorem print as Theorem 7.1.6, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:isntype-mono = 7.1.6).

*Editorial note.* No context span and no external reference are needed: the statement and proof use no label from outside the retained range.

*Editorial note.* The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

*Editorial scope.* One original result of the pinned HoTT Book source tree, counted once: an embedding into an n-type has an n-type domain (author label thm:isntype-mono, book Theorem 7.1.6).

*Allowed standard prerequisites (editorial).*

- Embedding. A map f : X to Y is an embedding when for all x, x prime the action on paths ap(f) is an equivalence of the path types.. It is the hypothesis on the map.
- n-type. Y is an n-type when its path types are (n minus one)-types, with the level shifted down once per path iteration.. It is the level hypothesis on the codomain and the conclusion for the domain.
- Level of an equivalent type. A type equivalent to an (n minus one)-type is again an (n minus one)-type.. It transfers the level from the path type in Y to the path type in X.

**Theorem 7.1.6.** *If $f : X \to Y$ is an embedding and $Y$ is an $n$-type for some $n \geq -1$, then so is $X$.*

*Proof.* Let $x, x' : X$; we must show that $x =_X x'$ is an $(n-1)$-type. But since $f$ is an embedding, we have $(x =_X x') \simeq (f(x) =_Y f(x'))$, and the latter is an $(n-1)$-type by assumption. □